From quantum time slices to relativistic and Newtonian motion

Result, 2026-10-27. The finite-slice construction of Newton motion extends to a nonquadratic kinetic energy $T(P)$ whenever T' and V' are globally Lipschitz. The sharper mean rate stated below also uses the bounded third derivatives in §1. For the positive-energy relativistic dispersion

$$T_c(p) = \sqrt{m^2c^4 + c^2p^2} - mc^2, \qquad m, c > 0, \tag{1}$$

the constants in the packet estimates can be chosen independently of $c \geq c_0 > 0$. For fixed initial data and finite time, the finite-product coherent-state means obey

$$\sup_{0\leq t\leq T} \|z_{\hbar,\pi,c}(t) - z_{\mathrm{N}}(t)\|_\gamma \leq C_T \left(\hbar + |\pi| + \frac{1}{c^2}\right). \tag{2}$$

Here z_{N} is the Newton solution for the same V, π is an arbitrary positive time partition and $\gamma > 0$ fixes a phase-space norm. This is a mean bound; the packet widths remain $O(\sqrt{\hbar})$. Thus $\hbar \downarrow 0$, time refinement and $c \to \infty$ may be taken jointly at the level of these means. Explicit constants and a proof are given below.

The exact Hamiltonians are self-adjoint and, at each fixed $\hbar$, their nonrelativistic limit is strong resolvent, hence strong unitary, convergence. A separate spectral-Dirichlet calculation gives uniform position convergence through a free hard-wall bounce, even when $\hbar \downarrow 0$ and $c \to \infty$ together. These are one-particle, positive-energy constructions with an external potential; they do not construct a Lorentz-covariant interacting field theory or supply a positive universal action constant.

The coherent-state classical limit and relativistic Feynman–Kac representations have established precedents; this note assembles a specific joint bound and makes no claim of priority. In particular, Carmona–Masters–Simon (abstract) treat relativistic Schrödinger operators through processes with stationary independent increments, and Hiroshima–Ichinose–Lőrinczi (abstract) give a subordinate-process Feynman–Kac formula in a more general spin and magnetic setting.

Targeted advisor check (GPT-6.1 Sol, 2026-10-27). The written packet, domain, strong-resolvent and spectral-wall arguments were checked with corrections applied. This is not independent peer review of the result or of the imported operator and probability theorems. The centered $O(\hbar)$ improvement is a subsequent written derivation, not part of that earlier check.

1. A general dispersion and its quantum realization

Work on $L^2(\mathbb{R})$ with $Qx = x$, $P = -i\hbar\, d/dx$, and $[Q, P] = i\hbar$. Assume for the full operator argument that real $T, V \in C^\infty(\mathbb{R})$ are bounded below and that all derivatives of order at least two are bounded. Add constants so that $T, V \geq 0$, which affects only a phase. Write

$$L = \|V''\|_\infty, \qquad M = \|T''\|_\infty. \tag{3}$$

For the sharper mean estimate also write $L_3 = \|V'''\|_\infty$ and $M_3 = \|T'''\|_\infty$, which are finite under the standing smoothness assumptions. The finite-product estimates alone need only C^1 functions with globally Lipschitz derivatives; the smoother class makes the domain argument explicit. Define on Schwartz space

$$H_\hbar = T(P) + V(Q), \qquad U_\hbar(t) = e^{-itH_\hbar/\hbar}. \tag{4}$$

To specify the domain, fix units for Q, P and put $N = 1 + Q^2 + P^2$. Since T, V grow at most quadratically, $\|H_\hbar f\| \leq C_\hbar \|Nf\|$ on $\mathcal{S}$. Moreover

$$[H_\hbar, N] = -i\hbar\{T'(P)Q + QT'(P)\} + i\hbar\{V'(Q)P + PV'(Q)\}, \tag{5}$$

and the linear growth of T', V' and boundedness of their derivatives give $\|[H_\hbar, N]f\| \le C_\hbar \|Nf\|$. Therefore

$$|\langle H_\hbar f, N^2 f\rangle - \langle N^2 f, H_\hbar f\rangle| \le 2C_\hbar \langle f, N^2 f\rangle. \tag{6}$$

Schwartz space is a core for N^2. Nelson's commutator theorem gives essential self-adjointness; the N^2 form estimate also makes the unitary group leave $D(N)$ invariant and act strongly continuously in its graph norm. For the cited energy-limited theorem use $G = N^2 - \inf\sigma(N^2) \ge 0$ and $E_0 = \inf\sigma(N^2)$; the relevant form domain remains $D(N)$ and (6) bounds $G + E_0$. These are standard imported operator results: see Energy-Limited Quantum Dynamics, Theorems 3.7–3.8 and Corollary B.2 (statements and hypotheses read). No nonlocal analogue of the local integration-by-parts graph identity from the earlier proof is assumed.

For $s \ge 0$ put

$$K_s = e^{-isV(Q)/\hbar}, \quad D_s = e^{-isT(P)/\hbar}, \quad S_s = D_s K_s. \tag{7}$$

On $D(N)$, $(S_s - I)f/s \to -iH_\hbar f/\hbar$ and the difference quotients are bounded from the N graph domain to L^2. The exact orbit of $f \in D(N)$ is compact in that graph norm on $[0, T]$; a telescoping sum therefore proves that products of S_s over arbitrary positive partitions converge strongly to $U_\hbar(t)$ as the mesh vanishes. Density and unitarity extend the result to every L^2 state. This is a fixed-$\hbar$ assertion, not uniform operator-norm control as $\hbar \downarrow 0$ or $c \to \infty$.

2. The finite-slice spread and mean bounds

Choose $\gamma > 0$, $h_0 > 0$, $0 < \hbar \le h_0$, and a coherent Gaussian with means (q_0, p_0) and initial standard deviations

$$\operatorname{sd} Q = \sqrt{\gamma\hbar/2}, \qquad \operatorname{sd} P = \sqrt{\hbar/(2\gamma)}. \tag{8}$$

These are the usual coherent widths for $\gamma = 1/(m\omega)$. Use

$$\|(q,p)\|_\gamma = |q| + \gamma|p|, \quad C = \sqrt{2\gamma}, \quad \beta = M/\gamma + \gamma L, \quad a = \max(M/\gamma, \gamma L). \tag{9}$$

When $a = 0$, interpret $(e^{aT} - 1)/a$ below as T.

For the state after each complete cell define $W_j = \operatorname{sd} Q_j + \gamma \operatorname{sd} P_j$. The exact finite-factor identities on Schwartz space are

$$K_s^* P K_s = P - sV'(Q), \qquad D_s^* Q D_s = Q + sT'(P). \tag{10}$$

Since the standard deviation of a Lipschitz function is at most its Lipschitz constant times the original standard deviation,

$$W_{j+1} \le (1 + sM/\gamma)(1 + s\gamma L) W_j, \qquad W_j \le C\sqrt{\hbar}\, e^{\beta T}. \tag{11}$$

The same uniform bound covers the intermediate post-kick state. The state need not remain Gaussian.

Let q_j, p_j be the packet means. The exact update reads

$$p_{j+1} = p_j - s\langle V'(Q)\rangle_j, \qquad q_{j+1} = q_j + s\langle T'(P)\rangle_{j,\mathrm{kick}}. \tag{12}$$

Introduce constants independent of the partition and of $0 < \hbar \le h_0$:

$$\begin{gathered} S = C\sqrt{h_0}\, e^{\beta T}, \qquad \bar f = |V'(0)| + LS, \qquad \bar u = |T'(0)| + (M/\gamma)S, \\ B = \gamma \bar f + \bar u, \qquad R = e^{\beta T}(\|(q_0, p_0)\|_\gamma + TB), \\ F_T = |V'(0)| + L(R + S), \qquad U_T = |T'(0)| + (M/\gamma)(R + S). \end{gathered} \tag{13}$$

Equation (12) and (11) give

$$\|(q_{j+1}, p_{j+1})\|_\gamma \le e^{\beta s}\|(q_j, p_j)\|_\gamma + sBe^{\beta s}; \tag{14}$$

hence all vertex means have norm at most R, $|\langle V'(Q)\rangle_j| \le F_T$, and $|\langle T'(P)\rangle_{j,\mathrm{kick}}| \le U_T$.

Linearly interpolate the means and define $\mathcal{F}_T(q,p) = (T'(p), -V'(q))$. Its Lipschitz constant in $\|\cdot\|_\gamma$ is at most a. On every cell,

$$\|\dot z_{\hbar,\pi} - \mathcal{F}_T(z_{\hbar,\pi})\|_\gamma \le C_\delta|\pi| + C_h\hbar, \qquad C_\delta = MF_T + \gamma LU_T, \quad C_h = \frac{C^2 e^{2\beta T}}{2}\left(\frac{M_3}{\gamma^2} + \gamma L_3\right). \tag{15}$$

For each mean, expand V' and T' around its corresponding current mean; the linear terms have zero expectation. Thus the force bias is at most $\frac{1}{2}L_3 \operatorname{Var}_j(Q)$, and the post-kick velocity bias is at most $\frac{1}{2}M_3 \operatorname{Var}_{j,\mathrm{kick}}(P)$. Use (11), including the intermediate-state bound, and add the within-cell changes $M|p_{j+1} - p(t)|$ and $L|q_j - q(t)|$. This proves (15) without assuming the propagated state is Gaussian.

Comparing two meshes and two positive values $\hbar, k$, integration and Gronwall give

$$\sup_{t\le T} \|z_{\hbar,\pi}(t) - z_{k,\sigma}(t)\|_\gamma \le G_T\{C_\delta(|\pi| + |\sigma|) + C_h(\hbar + k)\}, \quad G_T = \frac{e^{aT} - 1}{a}. \tag{16}$$

The interpolants are therefore Cauchy as both scales tend to zero, and their limit solves uniquely

$$\dot q = T'(p), \qquad \dot p = -V'(q). \tag{17}$$

No solution is assumed in defining that limit. At fixed $\hbar$ the strong product limit plus the uniform second-moment bounds recovers the exact unitary first moments: bounded spectral truncations pass by strong convergence and the tails are bounded by the second moments. Thus (15)–(16) also control the exact quantum means after the mesh is removed.

3. Relativistic dispersion and the Newton limit

For (1),

$$T_c'(p) = \frac{p}{m\sqrt{1 + p^2/(m^2c^2)}}, \qquad 0 < T_c''(p) = \frac{1}{m}\left(1 + \frac{p^2}{m^2c^2}\right)^{-3/2} \le \frac{1}{m}. \tag{18}$$

All hypotheses above hold for every $c > 0$, with common $M = 1/m$ and $T_c'(0) = 0$. For $c \ge c_0 > 0$,

$$|T_c'''(p)| = \frac{3|p|}{m^3c^2}\left(1 + \frac{p^2}{m^2c^2}\right)^{-5/2} \le \frac{3}{m^2c_0}. \tag{18a}$$

Consequently all constants in (11)–(16) can be chosen independently of $c \ge c_0$. The finite-c classical solution z_c from (17) is the positive-energy relativistic particle with

$$\dot q = \frac{pc^2}{\sqrt{m^2c^4 + c^2p^2}}, \qquad \dot p = -V'(q), \qquad |\dot q| < c. \tag{19}$$

The external potential is specified in a chosen frame; (19) does not assert full interacting Poincaré covariance.

The elementary inequality $1 - (1+x)^{-1/2} \le x/2$ for $x \ge 0$ gives

$$|T_c'(p) - p/m| \le \frac{|p|^3}{2m^3c^2}. \tag{20}$$

The Newton trajectory is already constructed by (17) with $T_\infty(p) = p^2/(2m)$. The common bound (13) gives $|p_{\mathrm{N}}(t)| \le R/\gamma$. Comparing its vector field with (19) and applying Gronwall,

$$\sup_{t\le T} \|z_c(t) - z_{\mathrm{N}}(t)\|_\gamma \le G_T \frac{(R/\gamma)^3}{2m^3c^2}. \tag{21}$$

Together with (15)–(16),

$$\sup_{t\le T}\|z_{\hbar,\pi,c}(t)-z_{\mathrm{N}}(t)\|_\gamma \le G_T\left[C_\delta|\pi|+C_h\hbar+\frac{(R/\gamma)^3}{2m^3c^2}\right]. \tag{22}$$

This proves (2). The bound is for fixed initial phase-space data and finite T; it is not uniform in momenta of order mc.

For any compactly supported smooth phase-space symbol b, the centered second-order Weyl Fourier argument in the groupoid companion converts the spread and mean bounds into

$$|\langle \mathrm{Op}_\hbar^W(b)\rangle_j - b(z_{\mathrm{N}}(t_j))| \le J_{b,\gamma}C^2e^{2\beta T}\hbar + I_{b,\gamma}G_T\left(C_\delta|\pi|+C_h\hbar+\frac{(R/\gamma)^3}{2m^3c^2}\right), \tag{23}$$

where

$$I_{b,\gamma}=(2\pi)^{-2}\int|\widehat{b}(\xi,\eta)|(|\xi|+|\eta|/\gamma)\,d\xi d\eta, \quad J_{b,\gamma}=\frac{1}{2(2\pi)^2}\int|\widehat{b}(\xi,\eta)|(|\xi|+|\eta|/\gamma)^2\,d\xi d\eta. \tag{24}$$

Thus a chosen diagonal schedule $\hbar = S_*\epsilon$, $|\pi| \le T_*\epsilon^\alpha$ and $c \ge c_*\epsilon^{-\kappa}$, with $\alpha,\kappa>0$, gives $O(\epsilon+\epsilon^\alpha+\epsilon^{2\kappa})$ for those observables at partition vertices. The groupoid deformation, mesh and c remain distinct until this schedule is chosen.

4. Fixed quantum action and the nonrelativistic limit

For fixed $\hbar>0$, define the nonnegative closed forms

$$q_c[\psi]=\|T_c(P)^{1/2}\psi\|^2+\|V(Q)^{1/2}\psi\|^2, \qquad D(q_c)=D(T_c(P)^{1/2})\cap D(V(Q)^{1/2}). \tag{25}$$

The functional calculus expression

$$T_c(p)=\frac{p^2/m}{1+\sqrt{1+p^2/(m^2c^2)}} \tag{26}$$

shows that $T_c(p)$ increases pointwise to $p^2/(2m)$ as $c\to\infty$. The increasing limit of the forms has domain $D(P)\cap D(\sqrt{V})$ and equals the nonrelativistic form. The monotone convergence theorem for closed forms therefore yields

$$H_{\hbar,c}\xrightarrow[c\to\infty]{\text{strong resolvent}}H_{\hbar,\infty}=P^2/(2m)+V(Q). \tag{27}$$

See Simon, "Tosio Kato's work," Theorem 10.12 (theorem statement read). Standard spectral convergence then gives strong convergence of the corresponding unitary groups, uniformly for t in compact intervals, for each fixed $\hbar$. This is not an operator-norm or uniform-in-$\hbar$ claim.

In Euclidean time the relativistic kinetic semigroup is positive, but its reference process is no longer Brownian in physical time. Indeed, for $a,b>0$ the subordination identity is

$$e^{-a\sqrt{b}}=\frac{a}{2\sqrt{\pi}}\int_0^\infty s^{-3/2}e^{-a^2/(4s)}e^{-bs}\,ds. \tag{28}$$

With $a=tc/\hbar$ and $b=p^2+m^2c^2$, multiply the right side of (28) by $e^{tmc^2/\hbar}$ to obtain $e^{-tT_c(P)/\hbar}$ as a positive mixture of Gaussian heat operators. Setting $p=0$ shows that the mixing measure has unit mass. The associated subordinate-Brownian process has jumps. Feynman–Kac with V is an additional standard construction, not a claim that the Gaussian midpoint variance or its large-field bound from the preceding note remains unchanged. For a general T with bounded T'', positivity of the Euclidean semigroup need not hold.

5. A relativistic spectral wall

The hard-wall model has to be specified again because $T_c(P)$ is nonlocal. Let Δ_D be the Dirichlet Laplacian on $(0,\infty)$ and define the **spectral** operator

$$H^D_{\hbar,c} = T_c\Big(\sqrt{-\hbar^2\Delta_D}\Big)\,. \tag{29}$$

It is self-adjoint. Under odd extension it is unitarily equivalent to $T_c(P)$ restricted to the odd subspace, since T_c is even.

Use the odd coherent state of the didactic guide, equation (20). Write its overlap as $\gamma_\hbar$, its initial full-line widths as σ_q, σ_p, and put

$$a_c(t) = q_0 + tT_c'(p_0), \qquad q_0 > 0, \quad p_0 < 0. \tag{30}$$

The full-line Heisenberg identity is $Q(t) = Q + tT_c'(P)$. For the evolved Gaussian $g_{\hbar,t}$,

$$\|(Q - a_c(t))g_{\hbar,t}\| \le \sigma_q + |t|\,\|[T_c'(P) - T_c'(p_0)]g_\hbar\| \le \sigma_q + \frac{|t|}{m}\sigma_p. \tag{31}$$

No preservation of Gaussian shape is needed. Apply $||x| - |a_c|| \le |x - a_c|$ to each of the two terms in the odd state. The half-line position expectation then satisfies

$$\Big|\langle Q\rangle^D_{\hbar,c}(t) - |a_c(t)|\Big| \le \sqrt{\frac{2}{1-\gamma_\hbar}}\left(\sigma_q + \frac{|t|}{m}\sigma_p\right). \tag{32}$$

Equation (20) also gives, uniformly for $|t| \le T$,

$$\left|\langle Q\rangle^D_{\hbar,c}(t) - \left|q_0 + \frac{p_0 t}{m}\right|\right| \le \sqrt{\frac{2}{1-\gamma_\hbar}}\left(\sigma_q + \frac{T}{m}\sigma_p\right) + \frac{T|p_0|^3}{2m^3c^2}. \tag{33}$$

The two classical impact times generally differ, but (33) is uniform through both. At fixed c, (32) gives the corresponding classical relativistic elastic bounce; at fixed q_0, p_0, T, (33) is $O(\sqrt{\hbar}+c^{-2})$ as the two limits are taken in any order.

The spectral operator in (29) subordinates the **killed Brownian** semigroup. It is distinct from first subordinating full-line Brownian motion and then killing the jump process when it exits the half-line. The latter gives a different nonlocal boundary problem; the odd extension above does not prove its hard-wall limit. See Kim–Song–Vondraček (introduction read) for this order-of-operations distinction. Nor is the local Schrödinger current in the guide automatically the probability current for this nonlocal Hamiltonian.

6. Consequence for STATE

For the positive-energy one-particle dispersion, bounded curvature closes the quantum packet Cauchy argument with constants uniform in c. The fixed-$\hbar$ form limit and the joint observable limit now connect relativistic and nonrelativistic mechanics in the same framework, with a separate spectral hard-wall example. The action unit is supplied, and no new Yang–Mills or positive-action-necessity conclusion follows.